



M-Tech Production & Marketing

Technology | Training | Marketing | Projects

TASK 8 REPORT

Internship Program — Batch 2026

Submitted By:	Ahmar Hayat
Department:	Web Dev
Submitted To:	Team lead
Date:	June 2026

1. Introduction

This report presents the work completed for Task 8 of the MTech Production & Marketing Web Development Internship Program, Batch 2026.

MTech Production & Marketing, based in Haripur, KPK, focuses on making websites interactive. HTML gives structure, CSS gives style, and now JavaScript gives behavior. This task is part of Phase 2: Frontend Development (Core) and covers JavaScript Essentials, DOM Manipulation, and ES6.

2. Objectives

The main objectives of this task were:

1. To understand what JavaScript is and how it works in the browser.
2. To learn JavaScript fundamentals: variables, data types, functions, loops, arrays, objects.
3. To learn ES6 modern syntax: let/const, arrow functions, template literals.
4. To learn DOM Manipulation how to make HTML dynamic.
5. To practice JavaScript on HackerRank, Exercism, and JavaScript30.

3. Work Done

1. Watched Video: "JavaScript Full Course" by freeCodeCamp 8 hours.
2. Watched Video: "JavaScript Crash Course" by Traversy Media.
3. Watched Video: "JavaScript DOM Manipulation" by Web Dev Simplified.
4. Watched Video: "ES6 JavaScript Tutorial" by The Net Ninja.
5. Used `document.getElementById()` and `addEventListener()` to make a button work.

4. Challenges Faced

1. DOM was confusing: I didn't understand `document.querySelector` at first. The Web Dev Simplified video made it clear: JS can "grab" HTML elements and change them.
2. ES6 Syntax: `const` vs `let` vs `var` was confusing. I learned `const` for values that don't change, `let` for values that do.

5. Key Learnings

1. What is JavaScript: JS is the programming language of the web. It makes websites interactive. Without JS, websites are just static pages.
2. JavaScript Fundamentals:
Variables: `let name = "Ahad", const age = 20`
Data Types: string, number, boolean, array, object
Functions: `function greet() {}` and ES6 `const greet = () => {}`

Loops: for, while

Arrays & Objects: How we store and manage data

3. ES6 Modern JavaScript:

let and const instead of var

Arrow Functions: `const add = (a, b) => a + b`

Template Literals: `Hello ${name}`

Destructuring: `const {title} = book`

4. DOM Manipulation: DOM = Document Object Model. It's how JS talks to HTML.

```
const btn = document.getElementById("myBtn");  
btn.addEventListener("click", () => {  
  alert("Hello from MTech!");  
});
```

With this we can change text, add elements, hide/show things.

5. Practice Platforms:

HackerRank: Good for logic and problem solving.

JavaScript30: Best for real projects. I built a working clock with just JS.

MDN + javascript.info: The 2 best docs for when I get stuck.

6. Conclusion

This was the hardest but most important lab so far. HTML and CSS were about looks. JavaScript is about logic.

ES6 is what everyone uses now. The old var and function syntax is outdated. Companies expect you to know arrow functions and template literals.

M-Tech Task Manager

A simple to-do list where you can add tasks, delete tasks, and it saves to browser. Uses everything from the lab: variables, functions, arrays, objects, DOM, ES6, events.

